Lab Inventory tool

- Internal use in Dow Lab
  - 4 location – same database

- Category

  A1. Chemicals/Reagents
    - (Peroxide, Regulatory related)

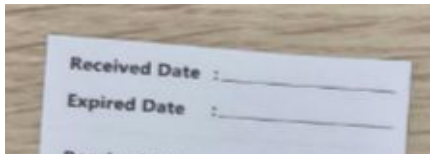
  A2. Calibration STD (verify STD)

  B. Gas

  C . Material supply, Consumable materials (some items that needs to control/monitor)

- Desired functions
  - Ordering system (user & focal points communication)
  - Inventory (amount) of Chemicals
    - min Stock and expired date (and extend expired date)
    - Chemical composition testing per criteria (Peroxide)
  - Status Dashboard (order status/chemical in stock status)

Basic concept



1. Order to cart

2. Sent email to focal point

3. Deliver & Check in

A: Print QR & Received date & Expired date (extend shelf life process)

4. Daily check Dashboard for min stock, expired & Peroxide monitoring

Peroxide monitoring (input result monitoring)

5. Check out

Empty bottle – dispose from inventory

Daily check dashboard

ข้อมูลของแต่ละหมู่ห้าLab, Location

User, Location

Location
Order
Checkin
Checkout
Peroxide
Transaction history
Extend shelflife
Regulatory related

Order status

Expired & Quarantine chemical

Min stock

Peroxide monitor

Functions

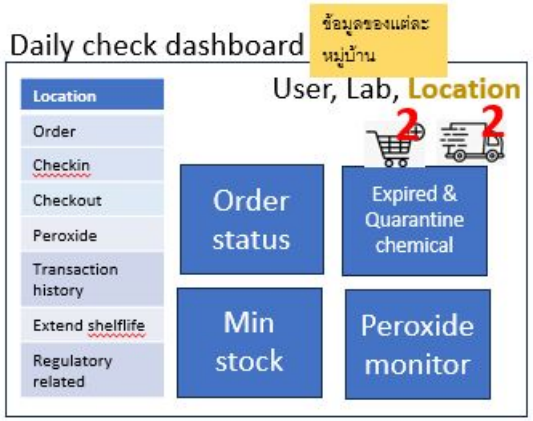
	Category	Ordering	Check-in	Notify/Remind	Check-out
A.1	Chemical & Reagent -Peroxide, Regulatory	x	x	x (stock/expired)	x
A.2	~~Calibration STD =~~ Verify STD	-	x	x (expired)	x
B	Gas	x	-	-	-
C	Material & consumable (only some item)	x	-	-	-

Order status	Stock status	Remind shelf life
Pending to order	In stock	-Near to expire
Waiting for delivery	-Min stock	-Peroxide monitor
	Out of stock	-Peroxide warning
		Expired (=out of stock)

Database structure

- 4 หมู่บ้าน – (AIE, MTP, CT, ATC)
  - **แต่ละหมู่บ้าน** มีห้องเก็บของของตัวเอง (Inventory per l
  - มี**ตัวแทนหมู่บ้าน** ที่ดูแลแต่ละเรื่อง (user: focal point)
    - หมู่บ้าน AIE มี 6 บ้าน (user: user)
    - หมู่บ้าน MTP มี 8 บ้าน
    - หมู่บ้าน CT มี 1 บ้าน
    - หมู่บ้าน ATC มี 1 บ้าน
  - **แต่ละการสั่งซื้อ** มีเลขสั่งซื้อ (PO)
- ใช้รายการของ (inventory) อันเดียวกัน

Location	Lab name
AIE	PO Lab
AIE	EOU Lab
AIE	PG Lab
AIE	POL Lab
AIE	SE Lab
AIE	Shared
MTP	EBSM Lab
MTP	PS Lab
MTP	Latex Lab
MTP	PU Lab
MTP	FM Lab
MTP	EFF Lab
MTP	PE Lab
MTP	Shared
CT	CT Lab
ATC	ATC Lab



PO Lab ซื้อสารเคมีจาก vendor ชื่อ ดินสอ = PO เบอร์ 12345
PO Lab ซื้อสารเคมีจาก vendor ชื่อ ยางลบ = PO เบอร์ 67890
EOU Lab ซื้อสารเคมีจาก vendor ชื่อ ดินสอ = PO เบอร์ 11223
EOU Lab ซื้อสารเคมีจาก vendor ชื่อ ยางลบ = PO เบอร์ 44556

1

- Location
- Order
- Checkin
- Checkout
- Peroxide
- Transaction history
- Extend shelflife
- Regulatory related

User, Lab, **Location**

Order stat

Dash ard2

ข้อมูลของแต่ละหมู่บ้าน

Dash board3

Dash board4

User, Lab, **Location**

Order

Search Item: Methanol

ตะกร้าหมู่บ้าน

Item	Quantity
Methanol AR	2
Methanol HPLC	1

User เข้ามาเลือกสั่งของ

User, Lab, **Location**

Order | Cart

3

No.	Status	Category	Item	Quantity	Vendor	Lab	PO	DateTime
1	Pending to order	Chemical & Reagent	Methanol AR	2	Chemex	PO lab	12345	15-Jan-26
2	Pending to order	Chemical & Reagent	Methanol HPLC	1	Chemex	PO lab	12345	15-Jan-26
3	Pending to order	Gas	N2 UHP	4	BIG	PO lab	67890	15-Jan-26

Approve order

Focal point เข้ามา approve ของที่สั่ง เลือก approve item ได้

Email sent to xxx, xxx, xxx, xxxx

Chemex
Item & Quantity & PO

BIG
Item & Quantity & PO

Focal point, Lab, **Location**

Order | Cart

No.	Status	Category	Item	Quantity	Vendor	Lab	PO	DateTime
1	Pending delivery	Chemical & Reagent	Methanol AR	4	Chemex	PO lab	12345	15-Jan-26
2	Pending delivery	Chemical & Reagent	Methanol HPLC	1	Chemex	PO lab	12345	15-Jan-26
3	Pending delivery	Gas	N2 UHP	4	BIG	PO lab	67890	15-Jan-26

Modify order

Approve order

หากต้องแก้ไข Focal point เข้า modify order และ approve order อีกครั้ง

Transaction history

บันทึก record ทุก transaction หรือ การ save

No.	Status	Category	Item	Quantity	Vendor	Lab	PO	DateTime
1	Pending delivery	Chemical & Reagent	Methanol AR	4	Chemex	PO lab	12345	15-Jan-26
2	Pending delivery	Chemical & Reagent	Methanol HPLC	1	Chemex	PO lab	12345	15-Jan-26
3	Pending delivery	Gas	N2 UHP	4	BIG	PO lab	67890	15-Jan-26

ข้อมูลของแต่ละหมู่บ้าน

Order status dashboard = 50items / scroll down

No.	Status	Category	Item	Quantity	Vendor	Lab		DateTime
1	Pending delivery	Chemical & Reagent	Methanol AR	4	Chemex	PO lab		15-Jan-26
2	Pending delivery	Chemical & Reagent	Methanol HPLC	1	Chemex	PO lab		15-Jan-26
3	Pending delivery	Gas	N2 UHP	4	BIG	PO lab		15-Jan-26
1	Pending to order	Chemical & Reagent	Methanol AR	2	Chemex	PO lab		15-Jan-26
2	Pending to order	Chemical & Reagent	Methanol HPLC	1	Chemex	PO lab		15-Jan-26
3	Pending to order	Gas	N2 UHP	4	BIG	PO lab		15-Jan-26

Approve order Date

Entry date

DASHBOARDS

2

Location
Order
Checkin
Checkout
Peroxide
Transaction history
Extend shelflife
Regulatory related

User, Lab, **Location**

2 2

Order status

Expired chemical

Min stock

Peroxide monitor

User, Lab, **Location**

Checkin – pending delivery

2 2

Item	Quantity	Lot*	Exp date
Methanol AR	2	METH123	31-Mar-30
Methanol HPLC	1	METH989	31-Mar-30

Checkin

User, Lab, **Location**

Transaction history

สรุปจำนวน label ที่ print

Pop up เด้งขึ้นมา แล้ว กด ok

CkIn
Open
OpenBy
EXP
Lot (see btl)*

User เข้ามารับของแยก line item ได้เลยไม่ต้องรวบการสั่งซื้อ

**ตัวอย่าง material supply ไม่ต้องขึ้น pending delivery

User, Lab, **Location**

Checkin - Manual

Item search

PO verify

Item	Quantity	Lot*	Exp date
PO verify STD1	1	POVER1	31-Mar-30
PO verify STD2	1	POVER2	31-Mar-30

Checkin

User, Lab, **Location**

Transaction history

No.	Status	Category	Item	Quantity	Vendor	Lab	PO	Checkin Date
1	In stock	Verify STD	PO Verify STD1	1	-	PO lab	-	20-Jan-26
2	In stock	Verify STD	PO Verify STD2	1	-	PO lab	-	20-Jan-26

ข้อมูลของแต่ละหมู่บ้าน

Min stock dashboard = 50items / scroll down

คำนวณ Total

No.	Reminder	Category	Item	Quantity	Min stock	Lab	Long Lead time
1	Min stock	Chemical & Reagent	Methanol AR	1	4	PO lab	30 days
2	Out of stock	Chemical & Reagent	Methanol HPLC	0	3	PO lab	-

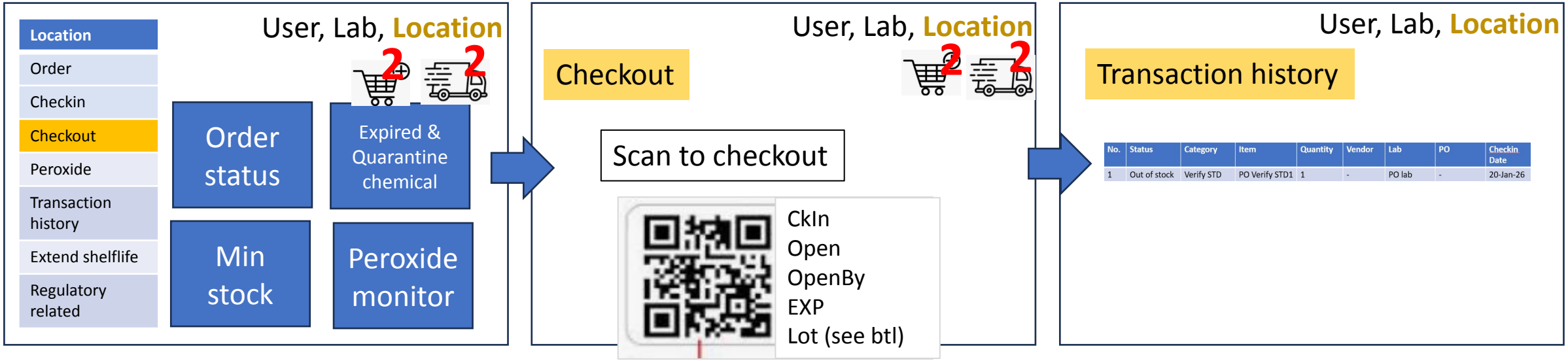
Expired date dashboard = 50items / scroll down

No.	Status	Category	Item	Days to expired	Expired date	Lab	Long Lead time
1	Near expire	Chemical & Reagent	Methanol AR	4 day	10-Mar-26	PO lab	-
2	In stock	Chemical & Reagent	Methanol HPLC	30 day	30-Mar-26	PO lab	30 days

นับวันย้อนหลังจากวันปัจจุบัน

DASHBOARDS

3



No.	Status	Category	Item	Quantity	Vendor	Lab	PO	Checkin Date
1	Out of stock	Verify STD	PO Verify STD1	1	-	PO lab	-	20-Jan-26

4

Location
- Order
- Checkin
- Checkout
- **Peroxide**
- Transaction history
- Extend shelf life
- Regulatory related

User, Lab, **Location**

2 2

Order status | Expired chemical

Min stock | Peroxide monitor

User, Lab, **Location**

Peroxide – all list of peroxide

Search: 2-Propanol

Ckln
Open
OpenBy
EXP
Lot (see btl)

No.	Reminder	Category	Item	Lot	Monitor due in	Monitor date	Lab
1	Peroxide monitor	Chemical & Reagent	2-Propanol	PA111	4 day	10-Mar-26	PO lab
2	Peroxide monitor	Chemical & Reagent	1,4-Dioxane	DIX222	30 day	6-Apr-26	PO lab
3	Peroxide monitor	Verify STD	PO Verify STD1	POVSTD1	45 day	21-Apr-26	PO lab
4	Peroxide monitor	Retain	PG USP/EP	C81500X	45 day	21-Apr-26	PG lab

Focal point, Lab, **Location**

Peroxide - entry

Class	Item (short name)	Lot
C	2-Propanol	PA111

		Result	
Checkin date	30-jun-25		V1
Inspect date	2-Jan-26	Normal	V2
Open date	15-Jan-26		V3
Monitor#1	30-Jan-26	ND	V4
Monitor#2	30-Jun-26	3	V5

Save

Focal point, Lab, **Location**

Peroxide - entry

Class	Item	Lot
C	2-Propanol	PA111

		Result (ppm)	
Checkin date	30-jun-25		V1
Inspect date	2-Jan-26	Normal	V2
Open date	15-Jan-26		V3
Monitor#1	30-Jan-26	ND	V4
Monitor#2	30-Jun-26	30	V5

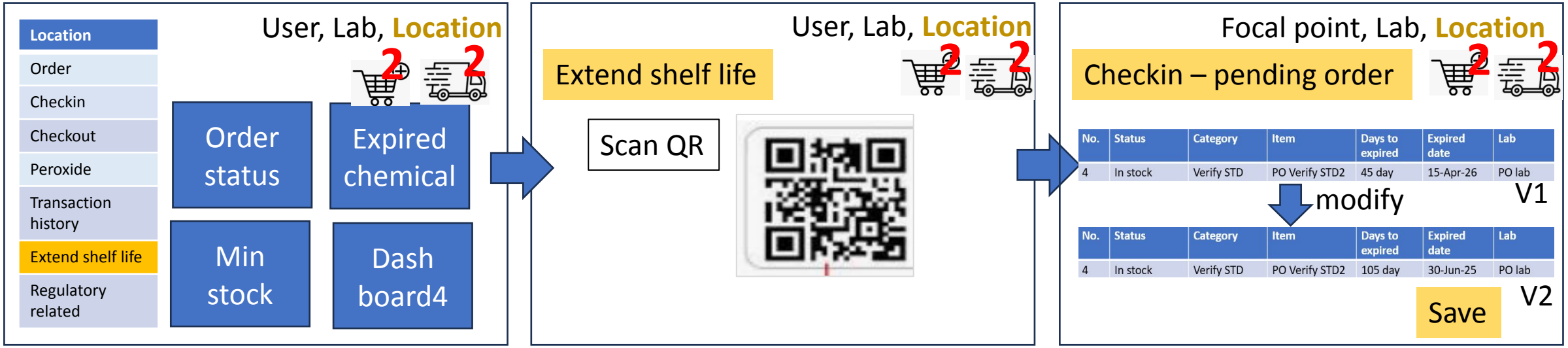
Save

Monitor result criteria

≥ 25ppm = Peroxide warning

> 100ppm = Quanrantine

5



Focal point เข้า modify expired date

ข้อมูลของแต่ละหมู่บ้าน

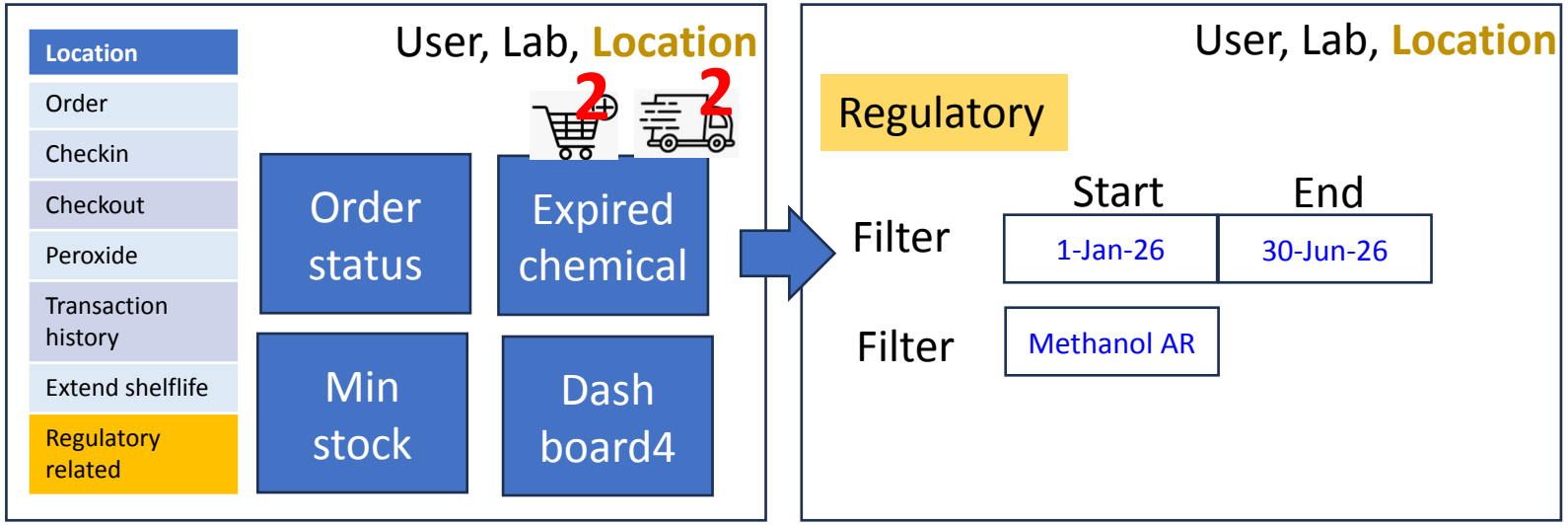
Expired dashboard = 50items / scroll down

No.	Reminder	Category	Item	Lot	Monitor due in	Monitor date	Lab
1	Peroxide monitor	Chemical & Reagent	2-Propanol	PA111	4 day	10-Mar-26	PO lab
2	Peroxide monitor	Chemical & Reagent	1,4-Dioxane	DIX222	30 day	6-Apr-26	PO lab
3	Peroxide monitor	Verify STD	PO Verify STD1	POVSTD1	45 day	21-Apr-26	PO lab
4	Peroxide monitor	Retain	PG USP/EP	C81500X	45 day	21-Apr-26	PG lab

นับวันย้อนหลังจากวันปัจจุบัน

DASHBOARDS

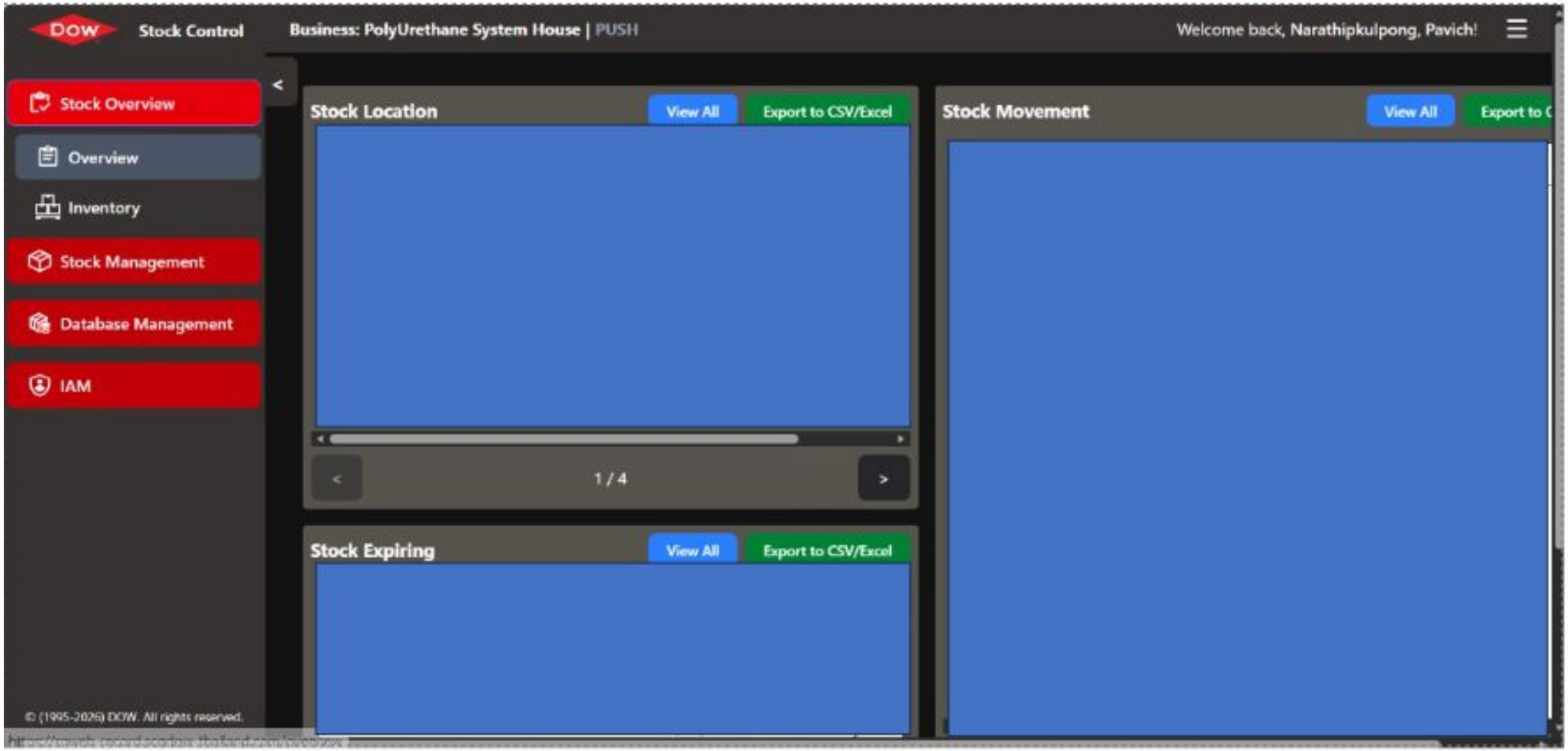
6



No.	Regulatory	Lab	Item	Bottle quantity	Quantity (L or kg)	Unit
1	VoOrOrKor7	PO lab	Methanol AR	4	10	L
2	VoOrOrKor7	DMC lab	Methanol HPLC	2	1	L

MaterialList

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
Category	Lab	Vendor	Part no.	Type	Regulatory	Lo	Item	Item (short name)	Size	Uni	Expired Date	Price 2026	long lead time	Total Min	AIE Min	MTP Min	CT Min	ATC Min
Chemical & Reagent	Shared	TCI	TCI-D2111-5G			NA	1,4-Di-tert-butylbenzene 1,4-Di-tert-butylbenzene "TCI" 5G		5	G		1,580		2	0	2	0	0
Chemical & Reagent	Shared	TCI	TCI-M0160-5G			NA	3-Methyl-2-benzothiazolinonehydrazone Hydrochloride "TCI" 5 G		5	G		2,290		1	1	0	0	0
Chemical & Reagent	Shared	TCI	TCI-P0101-500ML			NA	Phenol Red (0.04% in Water) "TCI" 500 ML		500	ML		3,430		4	4	0	0	0
Chemical & Reagent	Shared	TCI	TCI-P0782-5ML	Peroxide_B		NA	Propionaldehyde Dimethyl Acetal >98.0%(GC) "TCI" 5ML		5	ML		7,800		2	2	0	0	0
Chemical & Reagent	Shared	TCI	TCI-S0546-300G			NA	Sodium Bromide >99.5%(T) "TCI" 300 G		300	G		770		1	1	0	0	0
Chemical & Reagent	Shared	TCI	TCI-T0523-500ML			NA	Tripropylene Glycol (mixture of isomers) "TCI" 500 ML		500	ML		1,200		1	1	0	0	0
Chemical & Reagent	Shared	TCI	TCI-T0651-500G			NA	Tripropylene Glycol Monomethyl Ether "TCI" 500G		500	G		3,150		1	1	0	0	0
Chemical & Reagent	Shared	TRC	TRC-O239695-25G			NA	Octadecyl 3-?(3,?5-?Di-?tert-?butyl-?4-?hydroxyphenyl)?propionate (IR 1076) "TRC/Canada" 25 G		25	G		7,600		1	1	0	0	0
Chemical & Reagent	Shared	Accu	WC-OILG-10X-1			NA	OIL AND GREASE STANDARD 100 ML		100	L		6,700		2	2	0	0	0
Chemical & Reagent	Shared	Merck	114947.0001			NA	RECTANGULAR CELL 20 MM "Merck" 2Unit			EA		9,020		1	1	0	0	0
Chemical & Reagent	Shared	Sigma-Aldrich	SIA-240664-2L			NA	1,2-Dichlorobenzene anhydrous, 99% "Sigma-Aldrich" 2L			EA		13,200		4	0	4	0	0
Material & Consumable	Shared	S&T					Glass bottle 60 ML with septum and aluminum cap			EA								
Material & Consumable	Shared	S&T					Glass bottle 60 ML with septum and aluminium cap			EA								
Material & Consumable	Shared	S&T					Glass bottle 500 ML with septum and aluminium cap			EA								
Material & Consumable	Shared	S&T					Vial 2 mL clear color 9 mm neck โส คอเกลียว 100 PCS/PACK , KIMA			EA								
Material & Consumable	Shared	S&T					9mm write PTFE/red Silicone septa blue scew polypropylene cap 6 mm center hole 100 Pcs/Pack			EA								
Material & Consumable	Shared	S&T					Glass bottle 500 ML with septum and aluminium cap			EA								
Material & Consumable	Shared	TBS					Syringe NIPRO Luer lock 3 ml (100pcs/Box)			Box								
Material & Consumable	Shared	TBS					Syringe NIPRO Luer lock 3 ml (100pcs/Box)			Box								
Material & Consumable	Shared	TBS					NEEDLE NO.18G, LENGTH 1.5",(100PCS/BOX),			Box								
Material & Consumable	Shared	TBS					NEEDLE NO.18G, LENGTH 1.5",(100PCS/BOX),			Box								
Material & Consumable	Shared	TBS					Micro-Insert 250 ul.,31x5 mm. (100/PK) " Vertical		1	PK								
Material & Consumable	Shared	TBS					Micro-Insert 250 ul.,31x5 mm. (100/PK) " Vertical		1	PK								
Material & Consumable	Shared	TBS					Glass beaker size 100 ml			EA								
Material & Consumable	Shared	TBS					Glass beaker size 100 ml			EA								
Material & Consumable	Shared	TBS					Kimwipes EX-L Box - 280 Wipes (กระดาษเช็ดเลนส์)			Box								
Material & Consumable	Shared	TBS					Kimwipes EX-L Box - 280 Wipes (กระดาษเช็ดเลนส์)			Box								
Material & Consumable	Shared	TBS					Kimwipes EX-L Box - 280 Wipes (กระดาษเช็ดเลนส์)			Box								
Material & Consumable	Shared	TBS					Nalgene wash bottles for Methanol 500 ml			EA								
Material & Consumable	Shared	TBS					Nalgene wash bottles for Methanol 500 ml			EA								
Gas	Shared	BIG					Nitrogen UHP 99.999% Code:1100060			EA								
Gas	Shared	BIG					Argon UHP 99.999% Code: 1100033			EA								
Gas	Shared	BIG					Hydrogen UHP 99.999% Code:1100107			EA								
Gas	Shared	BIG					Helium Premier 99.999% Code: 1100201			EA								
Gas	Shared	BIG					Air Zero Code: 1100083			EA								
Gas	Sha																	
Chemical & Reagent	P																	
Verify STD	EB																	
Verify STD	EB																	
Verify STD	EB																	
Verify STD	EB																	

Database (example)

Example of Look & Feel